

Week 02 Face-to-Face Class Outline

Topic 1: IoT Application Enablers

1. The following technological advances enable realization and development of IoT:
 - a. **Portability:** Computing devices can be physically moved around easily
 - b. **Miniaturization:** Small form factors of computing devices enable easy embedding into physical spaces
 - c. **Low power & low heat:** Power efficiency of computing architectures and radio transceivers has been improved. The processors often support sleep mode. Energy harvesters are increasingly available. All these enable battery-based or even batteryless operations.
 - d. **Universal connectivity:** Wireless communications (Bluetooth, Wi-Fi, low-power wide-area radios) enable a larger scope of applications, anytime, anywhere.
 - e. **Artificial intelligence (AI):** Increasingly larger AI models enable IoT to deal with sophisticated problems.
 - f. **Convergence:** Smartphone is the largest converged platform/ecosystem that integrates music player, digital camera, gaming, etc.
 - g. **Divergence:** Specialized IoT devices are also developed and deployed for custom applications. Example: wearable health trackers.
2. Smartphone is a typical IoT device that meets the aspects a) to f) listed above.
3. Four key enabling technologies of IoT ecosystems
 - a. Sensors
 - b. Wireless communications
 - c. Cloud computing
 - d. AI

Topic 2: IoT Challenges

1. Reliability
 - a. Incident: Google Home global service outage on December 14th 2020
 - b. Centralized and decentralized control schemes exhibit different reliability profiles. Centralized control often offers better service quality but is subject to single-point failure risk. Decentralized control is free from single-point failures but harder to manage.
 - c. Centralized control with decentralized control as the fallback mechanism is probably a good design.
2. Security
 - a. Things are vulnerable to physical and cyber attacks

- b. Mirai attack (2016)
 - c. Ransom attacks may affect IoT systems in the future
- 3. Privacy
 - a. Privacy concerns due to things' operations in private space and time
 - b. Extended reading: <https://github.com/giladoved/webcam-heart-rate-monitor>
- 4. Big data
 - a. Massive data generated imposes processing and storage challenges
 - b. In 2025, nearly half of global data are generated by IoT devices
- 5. Interoperability
 - a. Diverse IoT devices, networks, and applications may not interoperate
 - b. Standardization for better interoperability is often an afterthought